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Government Litigation Risk and the Decline in Low-Income Mortgage Lending

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1 Big picture

- **Question.** Did government litigation risk push large lenders out of the Federal Housing Administration (FHA) mortgage market, and did that exit reduce mortgage credit access for low-income households?
- **Setting.** The paper studies U.S. FHA home purchase lending around the Department of Justice (DOJ) False Claims Act lawsuits against large FHA lenders in the 2010s.
- **Policy event.** Starting in 2012, DOJ/HUD litigation under the False Claims Act targeted lenders accused of fraudulent FHA lending practices. The authors identify **31 litigated lenders** and over **\$5 billion** in settlements.
- **Economic tension.** Enforcement can deter fraud and protect the public insurance fund, but large penalties and uncertain liability can also make regulated firms exit socially important markets.
- **Main empirical challenge.** Large banks may have reduced FHA lending for many reasons after the financial crisis, including bank regulation, changing local demand, or branch-network choices. The paper needs to isolate the effect of litigation risk.
- **Main designs.** The authors use:
 - lender-county-year difference-in-differences, comparing top FHA lenders to other lenders before and after 2012;
 - lender-county-year-loan-type triple differences, using conventional mortgages as a within-lender comparison market;
 - county-year difference-in-differences, exploiting counties’ pre-2012 exposure to large bank FHA lenders;
 - county-year-loan-type triple differences, again using conventional lending as a placebo/counterfactual.
- **Main lending result.** Large banks sharply exited the FHA market. At the lender level, top 5% bank lenders reduced FHA originations by roughly **58–60%**; at the county level, moving from no exposure to full exposure to large banks implies about a **17.8%** reduction in aggregate FHA lending.
- **Substitution result.** Other lenders, especially smaller shadow banks, partially filled the gap, but only about **60%** of the large-bank FHA volume loss was offset.
- **Quality result.** The contraction in FHA credit was **not** associated with meaningfully tighter underwriting, lower default risk, or better borrower experience. The representative loan officer in affected counties actually had higher misconduct rates.
- **Low-income access result.** High-exposure counties experienced a decline in overall mortgage

credit to low-income households. Moving from no exposure to full exposure lowers the low-income loan share by about **1.16 percentage points**, roughly **11%** of the 2010 mean.

- **Economic takeaway.** Legal enforcement can have unintended distributional costs: punishment aimed at large lenders' FHA practices reduced the supply of FHA credit in places where low-income borrowers relied on those lenders.

2 Data and measurement (key objects)

2.1 Institutional setting: FHA and the False Claims Act

- **FHA insurance.** The FHA insures mortgages made by participating lenders. Borrowers pay mortgage insurance premiums, and the insurance protects lenders against credit losses.
- **Borrower population.** FHA loans are especially important for first-time, lower-income, and lower-wealth borrowers because they allow low down payments. The current minimum down payment is **3.5%**, implying an LTV as high as **96.5%**.
- **Market importance.** During 2009–2017, FHA mortgages consistently represented more than **20%** of U.S. home purchase mortgages. For below-median-income homebuyers in 2010, FHA loans accounted for more than **50%** of the market.
- **False Claims Act risk.** FHA direct endorsement lenders certify that loans comply with FHA rules. The DOJ argued that violations connected to FHA insurance claims could constitute false claims, exposing lenders to large penalties.
- **Litigation wave.** A landmark 2012 Deutsche Bank/MortgageIT settlement was followed by a wave of DOJ/HUD investigations and settlements. The paper hand-collects settlement information from DOJ press releases and legal documents.
- **Why this matters.** A fine is not just a transfer if it changes future participation. If litigation risk makes large lenders leave FHA, the incidence can fall on borrowers who lose credit access.

2.2 Main data sources

- **HMDA data.** The Home Mortgage Disclosure Act data cover mortgage applications and originations. The paper uses HMDA to measure local FHA, conventional, and total low-income mortgage lending from **2009–2017**.
- **FHA administrative data.** Administrative HUD/FHA data cover the population of FHA-insured single-family mortgage originations from **2004–2017** and include underwriting variables and loan performance through September 2019.

- **NMLS loan officer data.** Nationwide Mortgage Licensing System data identify loan officers and misconduct records. The paper uses these data from **2012–2017** to measure the service quality of FHA lenders.
- **County controls.** The regressions also use county-year economic and demographic controls, including population, income, poverty, unemployment, education, minority share, house price growth, and average credit scores.

2.3 Sample definitions

- The main mortgage sample is restricted to **first-lien home purchase mortgages for owner-occupied one-to-four family dwellings**.
- **Banks** include depository institutions and mortgage banking affiliates inside bank holding companies.
- **Shadow banks** or **nonbanks** are non-depository independent mortgage companies.
- **Large lenders** are lenders in the **top 5%** of FHA home purchase origination volume in 2010.
- **Post** is an indicator for years **2012 and after**, because the major False Claims Act settlement wave begins in 2012.
- **Low-income borrowers** are defined following HUD as borrowers with incomes below **50%** of the FFIEC median family income in their county.

2.4 Key treatment and outcome variables

- **Top 5% lender treatment.**

$$\text{Top5}_i = 1\{\text{lender } i \text{ is in the top 5\% of FHA volume in 2010}\}.$$

This lender-level treatment captures both litigated lenders and large peers that may respond to the litigation risk.

- **County exposure.**

$$\text{Exposure}_c = \text{2010 FHA market share in county } c \text{ held by top-5\% bank lenders.}$$

A county with high exposure relied heavily on the large banks that later retreated from FHA lending.

- **Main quantity outcomes.**

– lender-county-year number of FHA originations;

- county-year log FHA dollar volume;
- FHA dollar volume per capita;
- FHA loans per 1,000 residents.
- **Quality outcomes.**
 - FICO score and indicator for $\text{FICO} \leq 620$;
 - debt-to-income ratio and indicator for $\text{DTI} \geq 50\%$;
 - early default within 12, 24, or 36 months;
 - mortgage interest rate;
 - county-level representative loan officer misconduct rate.

2.5 Important descriptive facts

- Low-income borrowers' share of U.S. home purchase mortgages fell from roughly **11%** in 2009 to about **6%** in 2017, with most of the decline coming through FHA lending.
- Nearly all False Claims Act settlements were concentrated among the very largest FHA lenders.
- The largest lenders were not obviously worse lenders before the litigation wave. In pre-2012 FHA administrative data, litigated lenders, top 5% lenders, and banks had lower early default rates than comparison lenders after controls.
- In the HMDA county-year sample, average FHA origination volume is **286 loans** per county-year, or **1.6 loans per 1,000 residents**; conventional lending is larger, with **573 loans** per county-year, or **3.5 loans per 1,000 residents**.
- Average county exposure to top 5% bank FHA lenders is **0.39**, with a standard deviation of **0.20**.

3 Models and empirical specifications (all equations + interpretation)

3.1 Pre-period default test: were targeted lenders worse?

The first descriptive regression tests whether lenders later targeted by DOJ litigation had unusually poor FHA loan performance before the litigation wave.

A simplified version is:

$$\text{EarlyDefault}_{\text{lict}} = \alpha + \beta_1 \text{Litigated}_i + \beta_2 \text{Top5}_i + \beta_3 \text{Bank}_i + X'_{\text{lict}} \theta + \gamma_{c,t} + \varepsilon_{\text{lict}},$$

where ℓ indexes loans, i lenders, c counties, and t origination years.

- EarlyDefault equals one if an FHA loan becomes 90 or more days delinquent within 24 months.
- X includes borrower and loan controls such as FICO, DTI, LTV, income, loan amount, first-time buyer status, gender, and minority status.
- $\gamma_{c,t}$ are county-by-year fixed effects.

Interpretation. If DOJ targeting were simply driven by poor loan performance, later litigated lenders should have higher pre-period early default rates. The paper instead finds lower or statistically indistinguishable default rates, weakening the idea that large banks were uniquely bad FHA underwriters before the lawsuits.

3.2 Lender-level difference-in-differences: did large lenders exit FHA?

The main lender-level design uses a Poisson model because the dependent variable is a count with many zeros:

$$E[Y_{ict} \mid \text{Top5}_i, \text{Post}_t] = \exp(\beta \text{Top5}_i \times \text{Post}_t + \delta_i + \gamma_{c,t}).$$

- Y_{ict} is the number of FHA home purchase mortgage originations by lender i in county c and year t .
- Top5_i identifies lenders in the top 5% of 2010 FHA volume.
- Post_t equals one for 2012–2017.
- δ_i are lender fixed effects.
- $\gamma_{c,t}$ are county-by-year fixed effects, absorbing local mortgage demand and economic conditions.

For a Poisson coefficient, the approximate percentage effect is:

$$\% \Delta Y = \exp(\hat{\beta}) - 1.$$

Interpretation. This compares the change in FHA lending by large lenders after 2012 to the change for other lenders operating in the same county-year. The county-year fixed effects are important because they net out local shocks to borrower demand for FHA loans.

3.3 Lender-level triple differences: is the effect FHA-specific?

To check whether large banks were merely retreating from all mortgage lending, the paper compares FHA lending with conventional lending in a lender-county-year-loan-type panel:

$$E[Y_{ictm} \mid \text{Top5}_i, \text{Post}_t, \text{FHA}_m] = \exp(\beta_1 \text{Top5}_i \times \text{Post}_t \times \text{FHA}_m \\ + \beta_2 \text{Top5}_i \times \text{Post}_t + \beta_3 \text{Top5}_i \times \text{FHA}_m \\ + \beta_4 \text{Post}_t \times \text{FHA}_m + \beta_5 \text{FHA}_m + \delta_i + \gamma_{c,t}).$$

- m indexes loan type: FHA versus conventional.
- FHA_m equals one for FHA observations.
- β_1 is the triple-difference coefficient.

Interpretation. The coefficient β_1 asks: after 2012, did large lenders reduce FHA lending more than conventional lending, relative to other lenders in the same county-year? A negative β_1 supports the interpretation that the shock is specific to FHA litigation risk rather than a general large-bank mortgage pullback.

3.4 County-level difference-in-differences: aggregate FHA credit supply

The lender-level results show that large banks reduced FHA lending, but they do not by themselves show whether aggregate credit supply fell, because other lenders may substitute. The county-level design is:

$$Y_{ct} = \beta_1 \text{Exposure}_c \times \text{Post}_t + \beta_2 \text{Controls}_{c,t-1} + \delta_c + \gamma_t + \varepsilon_{ct}.$$

- Y_{ct} measures aggregate FHA lending in county c and year t .
- Exposure_c is the 2010 FHA market share of top 5% bank lenders in the county.
- Controls include lagged county population, median household income, poverty, unemployment, education, minority share, house price growth, and average credit scores.
- δ_c and γ_t are county and year fixed effects.

Interpretation. This design asks whether counties that relied more heavily on large bank FHA lenders before the litigation wave experienced larger declines in total FHA lending after 2012.

3.5 County-level triple differences: FHA versus conventional markets

The county-level triple-difference specification compares FHA lending with conventional lending:

$$\begin{aligned} Y_{ctm} = & \beta_1 \text{Exposure}_c \times \text{Post}_t \times \text{FHA}_m + \beta_2 \text{Exposure}_c \times \text{Post}_t \\ & + \beta_3 \text{Exposure}_c \times \text{FHA}_m + \beta_4 \text{Post}_t \times \text{FHA}_m + \beta_5 \text{FHA}_m \\ & + \beta_6 \text{Controls}_{c,t-1} + \delta_c + \gamma_t + \varepsilon_{ctm}. \end{aligned}$$

Interpretation. This is the county-level analog of the lender-level triple difference. It asks whether high-exposure counties had a post-2012 decline in FHA lending relative to conventional lending. This helps rule out general county-level credit-demand shocks.

3.6 Substitution by other lenders

To study substitution, the paper decomposes county FHA originations into:

$$Y_{ct}^{\text{LargeBank}} \quad \text{and} \quad Y_{ct}^{\text{Other}}.$$

Then it estimates the county-level DiD separately for the two outcomes.

Interpretation. A large negative effect for large banks and a positive effect for other lenders means other lenders filled part of the gap. The key question is whether the positive response fully offsets the negative large-bank response.

3.7 Loan quality and borrower experience specifications

The quality tests use the same exposure-by-post logic, but with different dependent variables:

$$Q_{lct} = \beta_1 \text{Exposure}_c \times \text{Post}_t + X'_{lct} \theta + \delta_c + \gamma_t + \varepsilon_{lct},$$

where Q can be FICO, low-FICO indicator, DTI, high-DTI indicator, early default, or interest rate.

For loan officer misconduct, the unit is county-year:

$$\text{MisconductRate}_{ct} = \sum_i s_{ict}^{\text{FHA}} \cdot \text{MisconductRate}_i,$$

where s_{ict}^{FHA} is lender i 's FHA market share in county c and year t .

Interpretation. These tests ask whether litigation risk improved FHA loan quality. The central result is no: affected counties did not experience clearly better underwriting, lower default, lower mortgage rates, or better loan-officer quality.

3.8 Low-income mortgage access

The final main outcome is the low-income share of all mortgage lending:

$$\text{LowIncomeShare}_{ct} = \frac{\text{mortgage loans to low-income borrowers in county } c, t}{\text{all mortgage loans in county } c, t}.$$

The paper estimates the county-level DiD using this outcome, and then interacts exposure with rural/underserved and high-minority indicators.

Interpretation. This is the welfare-relevant credit-access outcome. If borrowers can simply substitute to conventional or other government loans, the total low-income share should not fall. The paper finds that it does fall, especially in rural and underserved counties.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard to identify

- The lawsuits were not randomly assigned across lenders.
- The post-2012 period also includes other changes in the mortgage market and bank regulation.
- Large banks may have been retreating from mortgage lending more generally after the financial crisis.
- Counties with high large-bank FHA shares may differ systematically from counties with low large-bank FHA shares.
- A fall in FHA lending might reflect lower borrower demand, not lower lender supply.

4.2 What identifies the lender-level effect

- The lender-level DiD compares top 5% FHA lenders with other lenders operating in the same county-year.
- County-by-year fixed effects absorb local economic conditions and borrower demand.
- Lender fixed effects absorb persistent lender-level differences.
- The treatment definition is grounded in the data: litigation settlements were concentrated among the top 5% of FHA lenders.
- The triple-difference version compares FHA lending with conventional lending, so it checks whether the large-lender contraction was FHA-specific.

4.3 What identifies the county-level effect

- County exposure is predetermined: it is measured using 2010 FHA market shares, before the main 2012 litigation wave.
- The identifying assumption is parallel trends: absent litigation, high- and low-exposure counties would have had similar FHA lending trends.
- The paper supports this assumption with dynamic event-study-style figures showing no meaningful pre-trends.
- Conventional mortgage lending provides a placebo/counterfactual market. High-exposure counties do not show a comparable post-2012 decline in conventional lending.
- Observable county characteristics are only weakly correlated with exposure, and regressions include rich county controls plus county and year fixed effects.

4.4 Why the evidence points to supply rather than demand

- The main shock is tied to lender legal risk, not borrower demand.
- The lender-level results condition on county-year fixed effects, comparing lenders facing the same local borrower demand.
- The county-level FHA decline does not appear in the conventional mortgage market.
- The FHA approval rate does not fall in high-exposure counties; instead, exiting lenders appear to stop taking FHA applications.
- Other lenders increase FHA lending in exposed counties, which is consistent with a supply gap left by large banks.

4.5 Why the paper studies top 5% lenders, not only litigated lenders

- Litigation can have a deterrent or “sobering” effect on peer firms that are not directly punished.
- Large non-litigated bank peers also reduced FHA lending after 2012.
- Using top 5% lenders captures the broader market-wide increase in perceived litigation risk.
- Robustness tests using only litigated lenders or top 10% lenders lead to similar conclusions.

4.6 Limitations

- The paper cannot directly observe the underlying fraud allegations or true fraud rates.
- Early default is an imperfect proxy for fraudulent lending.

- The design is strongest for measuring changes in observed mortgage supply, not for computing a full welfare calculation.
- The quality tests cannot rule out a uniform nationwide tightening of FHA underwriting by all lenders, because the main design is cross-sectional across high- and low-exposure counties.

5 Tables and figures

5.1 Figure 1: share of mortgages to low-income borrowers

Read:

- Figure 1 is the motivating fact.
- The share of U.S. home purchase mortgages going to low-income borrowers falls from roughly **11%** in 2009 to about **6%** in 2017.
- Panel A shows that most of the decline comes from FHA lending.
- Panel B shows that the decline is driven by large banks.
- This sets up the paper’s question: why did large banks retreat from a market central to low-income mortgage access?

5.2 Figure 2: FHA’s role in the mortgage market

Read:

- Figure 2 shows that FHA is not a small niche program.
- FHA loans account for more than **20%** of home purchase mortgages over 2009–2017.
- FHA loans are especially important for lower-income borrowers: among below-median-income homebuyers in 2010, FHA accounts for more than **50%** of originations.
- This makes the FHA market a natural place to study whether legal risk can affect credit access for financially constrained households.

5.3 Table 1 and Figures 3–4: the litigation wave and who was targeted

Read:

- Table 1 lists 31 litigated FHA lenders and settlement amounts totaling over **\$5 billion**.

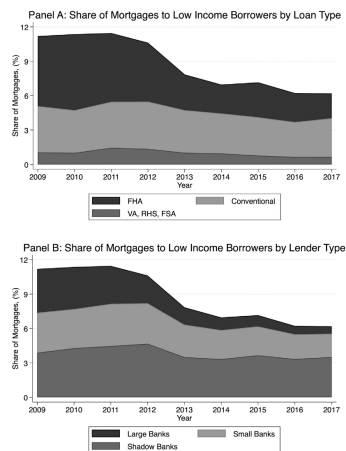


Fig. 1. Share of mortgages to low-income borrowers.
 Note: This figure shows the annual share of all U.S. home purchase mortgages that went to low-income borrowers from 2009 to 2017. Low-income borrowers are defined by HUD as those with incomes below 50% of the FHTEC median family income in their county. Panel A presents the low-income share decomposed by loan type: Federal Housing Administration loans, conventional loans, or all others (Veterans Administration, Rural Housing Service, or Farm Service Agency). Panel B presents the low-income share decomposed by lender type: large banks, small banks, or shadow banks.

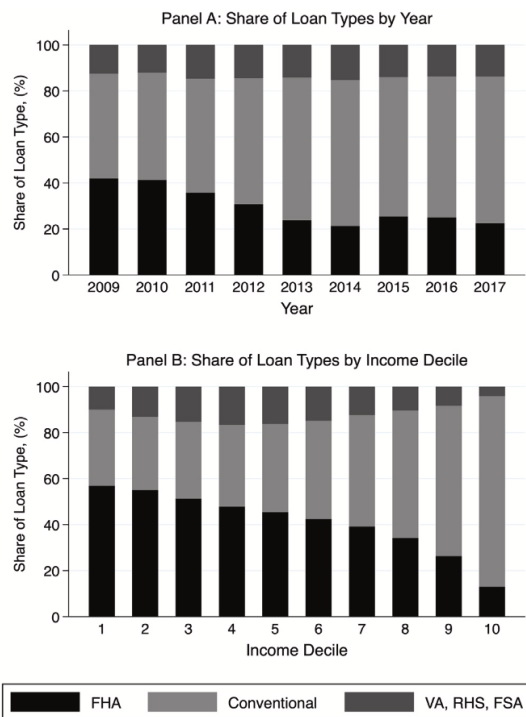


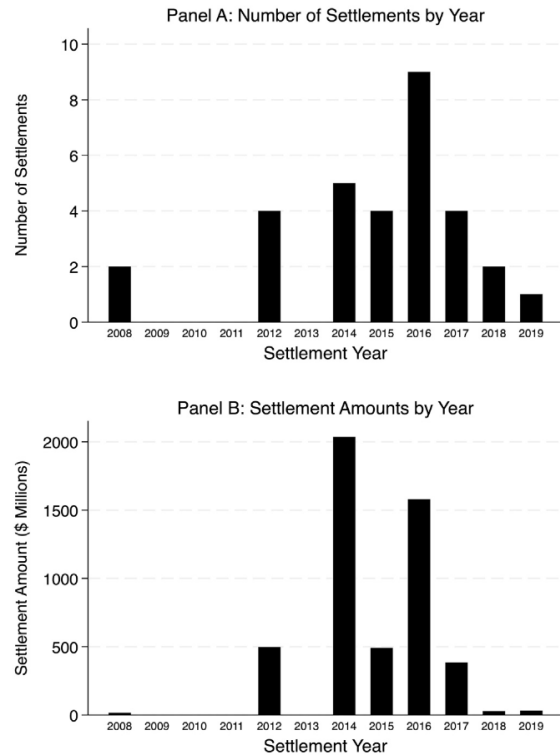
Fig. 2. Composition of the U.S. mortgage market.
 Note: This figure shows the composition of the U.S. home purchase mortgage market (by loan type) over time and across borrower income deciles. Panel A presents the share of home purchase mortgages from 2009 to 2017 by loan type: Federal Housing Administration loans, conventional loans, or all others (Veterans Administration, Rural Housing Service, or Farm Service Agency). Panel B presents a similar loan type decomposition across borrower income deciles using 2010 originations. The deciles are formed within states based on self-reported borrower incomes associated with originated loans.

Table 1

List of litigated FHA lenders.

Lender	Settlement date	Settlement amount (\$M)
Banks and Bank-Affiliates		
National City Mortgage Inc.	May. 2008	4.6
RBC Mortgage Company	Nov. 2008	10.7
Citimortgage, Inc. (Citibank)	Feb. 2012	158.3
Flagstar Bank	Feb. 2012	132.8
Deutsche Bank (MortgageIT)	May 2012	202.3
JPMorgan Chase	Feb. 2014	614.0
U.S. Bank	Jun. 2014	200.0
SunTrust Mortgage Inc.	Jun. 2014	418.0
Bank of America (Countrywide)	Aug. 2014	800.0
MetLife Home Loans LLC	Feb. 2015	123.5
First Tennessee Bank	Jun. 2015	212.5
Fifth Third Bancorp	Oct. 2015	84.9
Wells Fargo Bank	Apr. 2016	1,200.0
M&T Bank	May 2016	64.0
Regions Bank	Sep. 2016	52.4
BB&TC	Sep. 2016	83.0
IberiaBank	Dec. 2017	11.6
Shadow Banks		
Capmark Financial LLC	Feb. 2012	3.9
John Adams Mortgage Company	Dec. 2014	4.2
Franklin American Mort. Co.	Dec. 2015	70.0
Freedom Mortgage Corp.	Apr. 2016	113.0
Primary Residential Mortgage Inc.	Oct. 2016	5.0
SecurityNational Mort. Co.	Oct. 2016	4.3
MDR Mortgage Corp.	Nov. 2016	10.4
United Shore F.S. LLC	Dec. 2016	48.0
Prospect Mortgage, LLC	Jul. 2017	4.2
PHH	Aug. 2017	74.0
Allied Home Mortgage	Sep. 2017	296.0
Universal American Mort. Co. LLC	Oct. 2018	13.2
Gateway Funding	Dec. 2018	14.5
Quicken Loans	Jun. 2019	32.5

Note: This table lists the lenders that were litigated by the Department of Justice under the False Claims Act for alleged fraudulent lending activities in the FHA mortgage market. The settlement dates are the month and year in which each lender settled with the DOJ, ranging from 2008 to 2019. The settlement amounts (in \$ millions) are hand-collected from legal documents and DOJ press releases. See Internet Appendix A for further details on the False Claims Act litigation and settlements.

**Fig. 3.** False Claims Act settlements by year.

Note: This figure shows the timing of the settlements resulting from the wave of False Claims Act litigation brought by the DOJ and HUD against FHA mortgage lenders. Panel A presents settlement counts by year from 2008 to 2019. Panel B presents total settlement amounts (\$ millions) by year.

- The largest settlement listed is Wells Fargo at **\$1.2 billion**; other large bank settlements include Bank of America/Countrywide, JPMorgan Chase, SunTrust, Deutsche Bank/MortgageIT, and U.S. Bank.
- Figure 3 shows that settlements were small and infrequent before 2012, then rose sharply through 2017.
- Figure 4 shows that settlements were concentrated among the top 5% of FHA lenders by 2010 volume.
- Figure 4 also shows that these large lenders were not outliers in early default rates before the litigation wave.

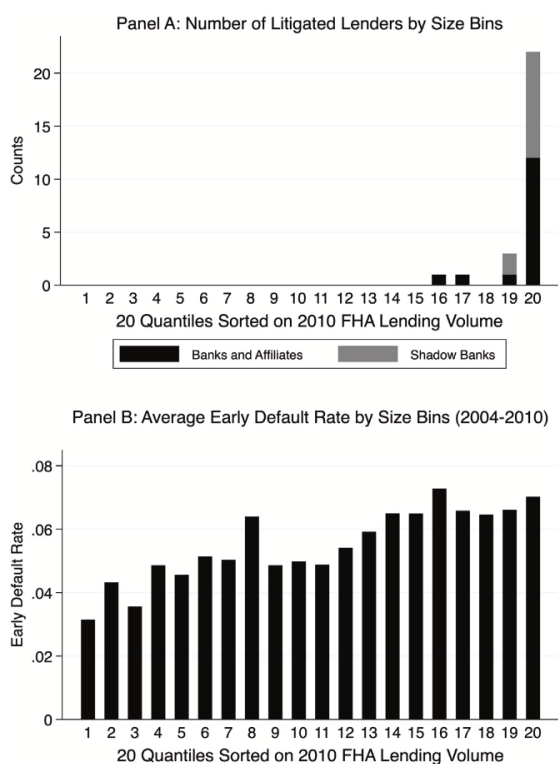


Fig. 4. Settlements and early default rates across FHA lender size bins. Note: This figure describes litigation activity and early default rates on FHA mortgages based on lender size. Panel A presents the number of litigated lenders that settled with the DOJ during 2008–2019, sorted across 20 quantiles based on FHA lending volume in 2010. The black shading represents banks and the gray shading represents shadow banks. We note that the following litigated lenders were not observed in the 2010 HMDA data: National City Mortgage, MortgageIT, Capmark Financial, and MDR Mortgage. Panel B presents the average early default rates of FHA lenders from 2004 to 2010, sorted across the same 20 quantiles of lender size. Lenders' early default rates are computed as the fraction of loans that became 90 or more days delinquent within two years of origination, based on administrative loan level data from the FHA.

Table 2
Analysis of FHA early mortgage default rates prior to the litigation wave.

	I(Early Default)			
	(1)	(2)	(3)	(4)
Litigated Lender	–0.007*** (0.002)			–0.002 (0.002)
Top 5% Lender		–0.006*** (0.002)		–0.002 (0.001)
Bank			–0.012*** (0.001)	–0.011*** (0.001)
Controls	Yes	Yes	Yes	Yes
County × Year FE	Yes	Yes	Yes	Yes
Adjusted R ²	0.120	0.120	0.120	0.120
# Loans	3,374,072	3,374,072	3,374,072	3,374,072

Note: This table presents OLS regressions examining early default rates on FHA loans originated prior to the DOJ lawsuits. The analysis explores whether default rates varied based on whether the originating lender was: (i) later litigated by the DOJ, (ii) a top 5% lender by 2010 FHA origination volume, and (iii) a bank. The analysis uses the population of Federal Housing Administration (FHA) home purchase loans originated from 2004 to 2010. The dependent variable, *I(Early Default)*, is an indicator for the loan becoming more than 90 days delinquent within 24 months of origination. The key independent variables are indicators for the lender later being litigated by the Department of Justice (*Litigated Lender*), for the lender's FHA volume being in the top five percent of all lenders in 2010 (*Top 5% Lender*), and for the lender being a depository institution (*Bank*). The control variables include narrow bins for FICO scores, DTI ratios, and LTV ratios, all of which are interacted with year of origination indicators. The controls also include the natural logs of income and loan amount, and indicators for first-time homebuyers, female borrowers, and minority borrowers, as well as county-year of origination fixed effects. Standard errors are clustered at the lender and county level and are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

5.4 Table 2: pre-litigation FHA default rates

Read:

Table 3

HMDA summary statistics.

Panel A: Lender-County-Year (N=2,543,000)					
	Mean	SD	P25	P50	P75
FHA Applications (#)	3.8	26.9	0	0	1
FHA Originations (#)	2.7	19.2	0	0	1
FHA Originations (\$ thousands)	490	3,925	0	0	80
Conventional Applications (#)	6.4	46.1	0	0	1
Conventional Originations (#)	4.7	34.4	0	0	1
Conventional Originations (\$ thousands)	1,293	14,103	0	0	214
Panel B: County-Year (N=23,820)					
	Mean	SD	P25	P50	P75
FHA Applications (#)	404	1,355	19	53	218
FHA Originations (#)	286	936	12	36	158
FHA Originations (Loans per 1,000 Residents)	1.6	1.2	0.7	1.2	2.1
FHA Originations (\$ Volume per Capita)	235	222	82	157	312
FHA Originations (ln(\$ Volume))	8.7	2.0	7.3	8.5	10.1
FHA Misconduct Rate 2012-2017 (%)	0.37	0.26	0.20	0.33	0.48
FHA Approval Rate (%)	71.7	12.0	66.7	73.3	78.6
Conventional Applications (#)	781	2,499	51	129	432
Conventional Originations (#)	573	1,810	32	89	318
Conventional Originations (Loans per 1,000 Residents)	3.5	2.4	1.8	2.9	4.5
Conventional Originations (\$ Volume per Capita)	644	686	225	412	787
Conventional Originations (ln(\$ Volume))	9.7	1.9	8.3	9.5	10.9
Conventional Misconduct Rate 2012-2017 (%)	0.15	0.13	0.06	0.12	0.21
Conventional Approval Rate (%)	73.8	9.8	68.8	75.4	80.3
FHA Share (%)	31.6	14.5	21.1	30.3	40.8
Low Income Loan Share (%)	8.5	6.3	4.0	7.5	11.8
Low Income Volume Share (%)	4.6	4.1	1.8	3.7	6.4
Exposure	0.39	0.20	0.25	0.37	0.51
Unemployment Rate _{t-1} (%)	7.3	2.9	5.2	6.9	9.0
Poverty Rate _{t-1} (%)	16.1	5.8	11.7	15.4	19.5
Minority Share _{t-1} (%)	20.7	18.1	6.2	14.0	31.4
Bachelor's Degree Share _{t-1} (%)	14.0	6.2	9.7	12.5	16.7
HPI Change _{t-1} (%)	0.03	5.29	-2.76	0.02	2.78
Population _{t-1} (thousands)	116	345	16	33	83
Median Income _{t-1} (\$ thousands)	46.5	11.7	38.6	44.4	51.7
Average Credit Score _{t-1}	673	26	653	674	692

Note: This table presents summary statistics describing FHA and conventional mortgage lending based on Home Mortgage Disclosure Act (HMDA) data. Panel A presents statistics on FHA and conventional mortgage applications and originations aggregated at the lender-county-year level from 2009 to 2017. We include only first-lien home purchase mortgages for owner-occupied one-to-four family dwellings. Lending is described in terms of both counts and dollar volumes (in thousands of dollars). Panel B presents similar statistics describing FHA and conventional lending at the county-year level, as well as per capita measures of lending volume and a range of socioeconomic characteristics. Internet Appendix Table A.1 provides variable definitions.

- Table 2 directly tests whether later-litigated lenders had worse pre-period FHA loan performance.
- The coefficients on Litigated Lender and Top 5% Lender are negative in simple specifications: about **-0.7 percentage points** and **-0.6 percentage points**, respectively.
- The coefficient on Bank is about **-1.2 percentage points**; with all indicators included, banks still have significantly lower early default rates.
- The result does not prove there was no fraud, but it shows that litigation was not simply targeted at lenders with unusually high early default rates.

5.5 Table 3: HMDA summary statistics

Read:

- The HMDA lender-county-year panel contains **2,543,000** observations.
- In the average lender-county-year, a lender receives **3.8** FHA applications and originates **2.7** FHA loans.
- The county-year panel contains **23,820** observations.
- The average county-year has **286** FHA originations and **573** conventional originations.
- Average county exposure to top 5% bank FHA lenders is **0.39**, with a standard deviation of **0.20**.

- This variation in exposure is the key source of identification for the county-level analysis.

5.6 Figure 5 and Table 4: large banks exit FHA

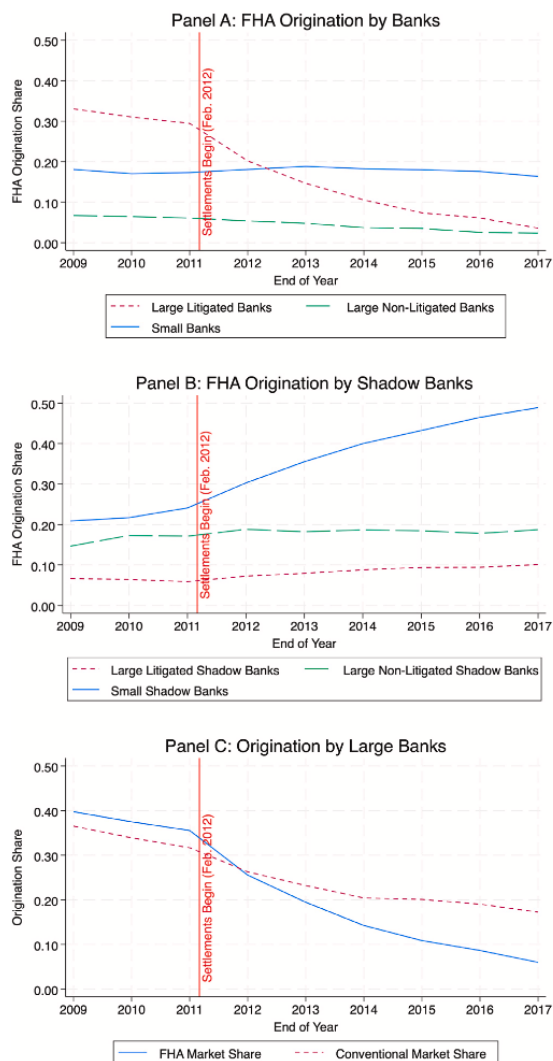


Fig. 5. Market shares by year.

Note: This figure displays the evolution of market shares in the home purchase mortgage market by lender type from 2009 to 2017. Panels A and B focus on the FHA market and classify lenders into six mutually exclusive groups, based on their status as banks versus shadow banks, and then their status as large litigated lenders versus large non-litigated lenders versus small lenders. Large lenders are those in the top five percent in terms of FHA lending volume in 2010. Panel A presents the market shares for the three groups of banks over time. Panel B presents the same breakdown for shadow banks. Panel C compares large banks' market share in the FHA versus conventional mortgage market over time. In all panels, the vertical line marks February 2012, the beginning of the wave of large False Claims Act settlements.

Table 4
Effect of the litigation wave on large FHA lenders.

Panel A: FHA Loan Originations	Number of Loans		
	All (1)	Banks (2)	Shadow Banks (3)
Top 5% Lender × Post	-0.458** (0.186)	-0.871*** (0.205)	-0.069 (0.123)
Lender FE	Yes	Yes	Yes
County × Year FE	Yes	Yes	Yes
2010 Mean of Top 5% Lenders	7.737	9.127	6.445
Pseudo R ²	0.517	0.554	0.521
# Observations	2,543,000	1,168,251	1,340,699
Panel B: FHA and Conventional Loan Originations	Number of Loans		
	All (1)	Banks (2)	Shadow Banks (3)
Top 5% Lender × Post × FHA	-0.370*** (0.088)	-0.494*** (0.124)	-0.333** (0.133)
Top 5% Lender × Post	-0.139 (0.158)	-0.392** (0.156)	0.267 (0.189)
Top 5% Lender × FHA	0.324*** (0.098)	0.556*** (0.083)	0.227 (0.142)
Post × FHA	-0.569*** (0.053)	-0.757*** (0.062)	-0.668*** (0.098)
FHA	-0.204*** (0.062)	-0.636*** (0.070)	0.358*** (0.112)
Lender FE	Yes	Yes	Yes
County × Year FE	Yes	Yes	Yes
2010 Mean of Top 5% Lenders	7.110	9.207	4.882
Pseudo R ²	0.508	0.534	0.523
# Observations	5,086,000	2,364,888	2,701,814

Note: This table presents Poisson regressions examining whether large FHA lenders responded to the increase in DOJ litigation risk in 2012. We examine mortgage lending activity from 2009 to 2017 using a lender-county-year panel constructed from HMDA data on first-lien home purchase mortgages for owner-occupied one-to-four family dwellings. The sample is restricted to: (i) lenders that originated FHA loans during the sample period, and (ii) lender-county-year pairs for which there was at least one mortgage application of any type. Panel A presents difference-in-differences tests examining the effect on FHA lending volume using the lender-county-year panel. The dependent variable, *Number of Loans*, is the number of FHA loan originations in the lender-county-year. The key independent variable is the interaction between *Top 5% Lender*, which is an indicator equal to one if the lender was in the top five percent when ranked by its FHA origination volume in 2010, and *Post*, an indicator for the year being 2012 or later. The specifications also include lender fixed effects and county-year fixed effects. Column 1 presents results for the full sample of lenders, column 2 focuses on the subsample of banks, and column 3 focuses on the subsample of shadow banks. Panel B presents triple-differences tests examining the same effects. These tests use a lender-county-year-loan type panel, which is expanded to include observations for both the lender's FHA volume and their conventional volume (which are measured with *Number of Loans* for the respective loan type). The key independent variable is the triple interaction between *Top 5% Lender*, *Post*, and the *FHA* indicator for the observation corresponding to FHA lending. Standard errors are clustered at the lender and county level and are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Read:

- Figure 5 shows the market-share transition visually.

- Litigated large banks' FHA market share falls from over **30%** in 2011 to less than **5%** in 2017.
- Large non-litigated banks also decline, from about **6%** to less than **3%**.
- Large shadow banks do not show a similar FHA retreat, while smaller shadow banks increase their market share from about **23%** in 2011 to just under **50%** by 2017.
- Table 4 Panel A estimates this formally. The top 5% lender coefficient is **-0.458**; for banks it is **-0.871**, implying roughly a **58%** decline in FHA originations by large banks.
- The corresponding shadow-bank coefficient is small and statistically insignificant in the FHA-only DiD.
- Table 4 Panel B uses conventional lending as a comparison market. The triple-difference coefficient for banks is **-0.494**, confirming that the retreat is specific to FHA lending.

5.7 Figures 6–7 and Table 5: aggregate county-level FHA lending falls

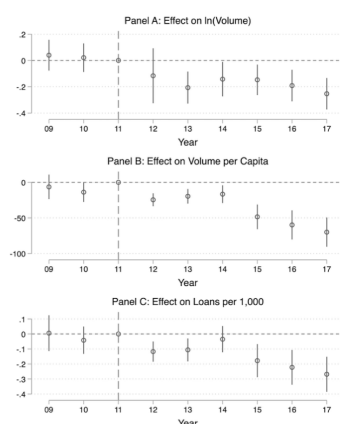


Fig. 6. Dynamic difference-in-differences estimates: County-year level FHA mortgages.
Note: This figure presents the dynamics of the main difference-in-differences results showing the effect of *Exposure* on FHA lending volume at the county-year level. *Exposure* is defined as the 2010 market share of banks in the top five percent of FHA origination volume. To produce the dynamic estimates, we regress measures of lending volume on the *Exposure* variable interacted with dummies for each year from 2009 to 2017 (with 2011 omitted), as well as the controls and fixed effects outlined in Table 5. Panels A, B, and C plot the estimated effects on FHA lending volume measured in terms of *ln(Volume)*, *Volume per Capita*, and *Loans per 1,000*, respectively. In each panel, the dashed vertical line marks the baseline year (2011) and the point estimates are accompanied by 95% confidence intervals.

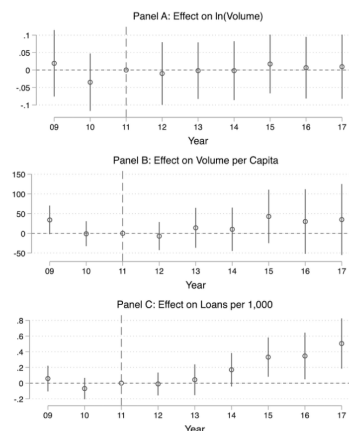


Fig. 7. Dynamic difference-in-differences estimates: County-year level conventional mortgages.
Note: This figure presents the dynamics of the (placebo) difference-in-differences results showing the effect of *Exposure* on conventional mortgage lending volume at the county-year level. *Exposure* is defined as the 2010 market share of banks in the top five percent of FHA origination volume. To produce the dynamic estimates, we regress measures of conventional lending volume on the *Exposure* variable interacted with dummies for each year from 2009 to 2017 (with 2011 omitted), as well as the controls and fixed effects outlined in Table 5. Panels A, B, and C plot the estimated effects on conventional lending volume measured in terms of *ln(Volume)*, *Volume per Capita*, and *Loans per 1,000*, respectively. In each panel, the dashed vertical line marks the baseline year (2011) and the point estimates are accompanied by 95% confidence intervals.

Read:

- Table 5 is the main aggregate credit-supply result.
- In Panel A, the exposure-by-post coefficient for log FHA volume is **-0.196**. Moving from zero exposure to full exposure implies about a **17.8%** decline in FHA lending.
- Because the standard deviation of exposure is 0.20, a one-standard-deviation increase in exposure implies about a **3.8%** decline in average FHA lending.
- Panel B's triple-difference coefficient is **-0.194**, implying about a **17.6%** FHA decline relative to conventional lending.
- Figure 6 shows no strong pre-trend before 2012 and a sharp negative effect beginning in 2012.

- Figure 7 shows no comparable negative effect for conventional mortgages, supporting the interpretation that the effect is FHA-specific.

Table 5
Aggregate effect of the litigation wave: county-year level evidence.

Panel A: FHA Loan Originations			
	ln(Volume) (1)	Volume per Capita (2)	Loans per 1,000 (3)
Exposure × Post	−0.196*** (0.039)	−32.837*** (5.818)	−0.141*** (0.042)
Controls	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
2010 Mean	8.784	246.385	1.709
Adjusted R ²	0.960	0.881	0.861
# Observations	23,820	23,820	23,820
Panel B: FHA and Conventional Loan Originations			
	ln(Volume) (1)	Volume per Capita (2)	Loans per 1,000 (3)
Exposure × Post × FHA	−0.194*** (0.046)	−43.351 (33.337)	−0.378*** (0.126)
Exposure × Post	0.003 (0.030)	10.109 (29.254)	0.235** (0.104)
Exposure × FHA	−0.370*** (0.064)	−123.112*** (20.201)	−0.755*** (0.099)
Post × FHA	−0.479*** (0.020)	−347.076*** (14.963)	−1.538*** (0.056)
FHA	−0.493*** (0.028)	−119.704*** (8.455)	−0.502*** (0.044)
Controls	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
2010 Mean	9.040	317.151	2.021
Adjusted R ²	0.950	0.678	0.732
# Observations	47,640	47,640	47,640

Note: This table presents OLS regressions examining the effects of the 2012 increase in DOJ litigation risk and large banks' exit from the FHA market on total FHA lending volumes. We examine mortgage lending activity from 2009 to 2017 using a county-year panel constructed from HMDA data on first-lien home purchase mortgages for owner-occupied one-to-four family dwellings. Panel A presents difference-in-differences tests using the county-year panel. The dependent variables in columns 1, 2, and 3, are *ln(Volume)*, the natural logarithm of one plus the dollar volume of FHA loan originations, *Volume per Capita*, the total dollar volume of FHA originations per capita, and *Loans per 1,000*, the total number of FHA originations per 1,000 residents in the county. The key independent variable is the interaction term *Exposure × Post*. *Exposure* is the FHA market share of large banks (those among the top 5% largest FHA lenders) as of 2010 in a given county. *Post* is an indicator for the year being 2012 or later. The specifications also include county-year level controls listed and defined in Internet Appendix Table A.1, as well as county fixed effects and year fixed effects. Panel B presents triple-differences tests examining the same effects. These tests use a county-year-loan type panel, which is expanded to include observations for both FHA volume and conventional volume in the county-year. The key independent variable is the triple interaction between *Exposure*, *Post*, and the *FHA* indicator for the observation corresponding to FHA lending. Standard errors are clustered at the county level and are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Table 6
Did other lenders (partially) substitute for exiting large banks?

FHA Loan Originations						
	Loan Volume (\$Millions)		Volume per Capita		Loans per 1,000	
	Large Banks (1)	Others (2)	Large Banks (3)	Others (4)	Large Banks (5)	Others (6)
Exposure × Post	−10.659*** (1.943)	6.052*** (1.725)	−69.439*** (4.742)	36.851*** (5.336)	−0.570*** (0.033)	0.425*** (0.033)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
2010 Mean	21.321	36.234	91.677	154.545	0.643	1.062
Adjusted R ²	0.783	0.956	0.699	0.861	0.697	0.850
# Observations	23,820	23,820	23,820	23,820	23,820	23,820

Note: This table presents difference-in-differences (OLS) regressions examining the extent to which other lenders substituted for large banks exiting the FHA market. We examine mortgage lending activity from 2009 to 2017 using a county-year panel constructed from HMDA data on first-lien home purchase mortgages for owner-occupied one-to-four family dwellings. The dependent variable in columns 1 and 2 is *Loan Volume (\$ Millions)*, which is the dollar volume of FHA loan originations in millions of dollars for the respective group of lenders (large banks in column 1 and all other lenders in column 2). The dependent variable in columns 3 and 4 is *Volume per Capita*, which is the total dollar volume of FHA originations per capita made by the respective group of lenders. The dependent variable in columns 5 and 6 is *Loans per 1,000*, which is the total number of FHA originations per 1,000 residents in the county made by the respective group of lenders. The key independent variable is the interaction term *Exposure × Post*. *Exposure* is the FHA market share of large banks (those among the top 5% largest FHA lenders) as of 2010 in a given county. *Post* is an indicator for the year being 2012 or later. The specifications also include county-year level controls listed and defined in Internet Appendix Table A.1, as well as county fixed effects and year fixed effects. Standard errors are clustered at the county level and are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

5.8 Table 6: other lenders partially substitute

Read:

- Table 6 decomposes FHA lending into large-bank lending versus all other lenders.
- In a fully exposed county, large banks reduce FHA volume by about **\$10.7 million** per year after 2012.
- Other lenders increase FHA volume by about **\$6.1 million** per year.
- Thus, other lenders replace only about **60%** of the large-bank reduction.

- Results are similar using volume per capita and loans per 1,000 residents.
- The key economic point is that entry/expansion by other lenders is real but incomplete.

5.9 Table 7, Table 8, and Table 9: loan characteristics, underwriting, and default risk

Table 7
Summary statistics for FHA mortgages.

Panel A: Full Sample					
	Mean	SD	P25	P50	P75
FICO	690	50	651	682	724
LTV	95.4	4.3	96.5	96.5	96.5
Mortgage Amount	176,867	90,343	113,898	157,102	220,873
Mortgage Rate	4.36	0.68	3.75	4.25	4.88
First-time Home Buyer	0.798	0.402	1	1	1
Borrower Age	37.1	11.2	28.0	34.0	44.0
Female	0.368	0.482	0	0	1
Married	0.502	0.500	0	1	1
White	0.674	0.469	0	1	1
Black	0.110	0.313	0	0	0
Asian	0.032	0.176	0	0	0
Hispanic	0.170	0.375	0	0	0
Borrower Income	64,878	34,202	40,032	56,556	80,796
Borrower Assets	19,769	33,355	6,859	11,398	20,758
DTI (back-end)	0.407	0.091	0.345	0.415	0.475
Default Rate (1-yr)	0.014	0.117	0	0	0
Default Rate (2-yr)	0.048	0.214	0	0	0
Default Rate (3-yr)	0.078	0.269	0	0	0
# Loans	5,277,476				

Panel B: By Lender Type						
	Large Banks		Small Banks		Shadow Banks	
	Mean	SD	Mean	SD	Mean	SD
FICO	695	53	691	49	687	49
LTV	95.2	4.7	95.4	4.1	95.4	4.3
Mortgage Amount	169,509	92,916	160,281	78,830	185,239	91,969
Mortgage Rate	4.50	0.69	4.24	0.70	4.35	0.66
First-time Home Buyer	0.777	0.416	0.793	0.405	0.807	0.395
Borrower Age	36.5	11.2	36.2	11.1	37.5	11.2
Female	0.368	0.482	0.373	0.484	0.366	0.482
Married	0.489	0.500	0.493	0.500	0.511	0.500
White	0.693	0.461	0.742	0.437	0.644	0.479
Black	0.110	0.313	0.110	0.313	0.111	0.314
Asian	0.038	0.190	0.025	0.155	0.033	0.177
Hispanic	0.146	0.353	0.108	0.310	0.200	0.400
Borrower Income	64,320	35,306	61,837	32,481	66,115	34,276
Borrower Assets	20,815	34,223	16,988	29,772	20,314	34,109
DTI (back-end)	0.397	0.092	0.397	0.092	0.414	0.090
Default Rate (1-yr)	0.012	0.107	0.013	0.112	0.015	0.122
Default Rate (2-yr)	0.041	0.199	0.044	0.205	0.052	0.223
Default Rate (3-yr)	0.070	0.255	0.071	0.257	0.084	0.278
# Loans	1,160,258		1,039,156		3,078,062	

Note: This table presents summary statistics describing FHA mortgages originated from 2009 to 2017 using comprehensive administrative data on the population of FHA-insured single-family home purchase mortgages. Panel A presents statistics for the full sample of loans. Panel B compares loan and borrower characteristics for FHA loans originated by three different types of lenders: large banks, small banks, and shadow banks. Internet Appendix Table A.1 provides variable definitions.

Table 8
Effect of litigation exposure on FHA lending standards.

Panel A: Credit Scores		
	FICO (1)	I(FICO ≤ 620) (2)
Exposure × Post	-2.020 (2.730)	-0.017 (0.013)
Controls	No	No
County FE	Yes	Yes
Year FE	Yes	Yes
Adjusted R ²	0.041	0.014
# Loans	5,195,445	5,195,445

Panel B: Debt-to-Income Ratios		
	DTI (1)	I(DTI ≥ 50%) (2)
Exposure × Post	-0.003 (0.004)	0.002 (0.014)
Controls	No	No
County FE	Yes	Yes
Year FE	Yes	Yes
Adjusted R ²	0.042	0.017
# Loans	5,195,445	5,195,445

Note: This table presents difference-in-differences (OLS) regressions examining the effect of the 2012 increase in DOJ litigation risk on lending standards in the FHA mortgage market. The tests use comprehensive loan-level administrative data on all FHA-insured single-family home purchase mortgages originated from 2009 to 2017. The key independent variable is the interaction term *Exposure × Post*. *Exposure* is defined as the FHA market share of large banks (those among the top 5% largest FHA lenders) as of 2010 in a given county, which captures exposure to litigation risk. *Post* is an indicator for the year being 2012 or later. The specifications also include county fixed effects and year of origination fixed effects. In Panel A, lending standards are measured based on credit scores on originated FHA loans. Column 1 presents results where the dependent variable is the borrower's FICO score, and column 2 presents results where the dependent variable is an indicator for the borrower having a FICO score below 620. In Panel B, lending standards are measured based on debt-to-income ratios on originated FHA loans. Column 1 presents results where the dependent variable is the borrower's DTI, and column 2 presents results where the dependent variable is an indicator for the borrower having a DTI above 50%. Standard errors are clustered at the lender and county level and are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Read:

- Table 7 summarizes the FHA administrative data. Average FICO is **690**, average LTV is **95.4%**, average DTI is **40.7%**, and about **80%** of borrowers are first-time homebuyers.
- FHA borrowers served by large banks, small banks, and shadow banks look broadly similar, although shadow banks originate the largest number of FHA loans in the sample.
- Table 8 finds no meaningful tightening of underwriting in high-exposure counties. The FICO coefficient is **-2.020** and insignificant; the low-FICO and DTI measures are also insignificant.
- Table 9 finds no significant reduction in early default risk at 12-, 24-, or 36-month horizons.
- These results are important because they undercut the argument that the loss of FHA credit was accompanied by improved credit quality.

Table 9

Effect of litigation exposure on FHA default risk.

<i>Dependent Variable = $I(Early\ Default)$</i>						
	12 Months		24 Months		36 Months	
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure $\times$ Post	-0.002 (0.002)	-0.002 (0.002)	-0.006 (0.005)	-0.004 (0.005)	-0.013 (0.008)	-0.011 (0.008)
Controls	No	Yes	No	Yes	No	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.003	0.016	0.007	0.042	0.010	0.060
# Loans	5,195,445	5,195,444	5,195,445	5,195,444	5,195,445	5,195,444

Note: This table presents difference-in-differences (OLS) regressions examining the effect of the 2012 increase in DOJ litigation risk on the level of default risk on newly-originated FHA loans. The tests use comprehensive loan-level administrative data on all FHA-insured single-family home purchase mortgages originated from 2009 to 2017. The dependent variable, $I(Early\ Default)$, is an indicator for the loan becoming more than 90 days delinquent within either 12, 24, or 36 months of origination in columns 1-2, 3-4, and 5-6, respectively. The key independent variable is the interaction term $Exposure \times Post$. $Exposure$ is defined as the FHA market share of large banks (those among the top 5% largest FHA lenders) as of 2010 in a given county, which captures exposure to litigation risk. $Post$ is an indicator for the year being 2012 or later. All specifications include county fixed effects and year of origination fixed effects. Columns 2, 4, and 6 add loan-level controls including narrow bins for FICO scores, DTI ratios, and LTV ratios, the natural logs of income and loan amount, and indicators for first-time homebuyers, female borrowers, and minority borrowers. Standard errors are clustered at the lender and county level and are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Table 10

Effect of litigation exposure on borrowers' FHA lending experience.

<i>Panel A: Mortgage Rates</i>			
	Full Sample (1)	High Competition (2)	Low Competition (3)
Exposure $\times$ Post	-0.060 (0.045)	-0.077 (0.051)	0.059** (0.027)
Controls	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Adjusted R ²	0.679	0.679	0.682
# Loans	5,195,444	4,841,956	353,488

<i>Panel B: Loan Officer Misconduct Rates</i>		
	Misconduct Rate 2012–2017, (%) (1)	Misconduct Rate 2012, (%) (2)
Exposure $\times$ Post	0.126*** (0.021)	0.106*** (0.025)
Controls	Yes	Yes
County FE	Yes	Yes
Year FE	Yes	Yes
2010 Mean	0.271	0.246
Adjusted R ²	0.361	0.266
# Observations	23,820	23,820

Note: This table presents difference-in-differences (OLS) regressions examining the effect of the 2012 increase in DOJ litigation risk on borrowers' experience in the FHA mortgage market in terms of the interest rates charged and the quality of service (based on loan officer misconduct rates). Panel A presents tests examining interest rates on newly-originated FHA loans. The tests use comprehensive loan-level administrative data on all FHA-insured single-family home purchase mortgages originated from 2009 to 2017. The dependent variable is the mortgage interest rate in percentage point units. The key independent variable is the interaction term $Exposure \times Post$. $Exposure$ is defined as the FHA market share of large banks (those among the top 5% largest FHA lenders) as of 2010 in a given county, which captures exposure to litigation risk. $Post$ is an indicator for the year being 2012 or later. Column 1 presents the results for the full sample. Columns 2 and 3 present results focusing on high- and low-competition counties, respectively. High-competition counties are those in the top tercile in terms of the number of non-treated (i.e., not top five percent) FHA lenders operating in the county in 2010. All other counties are defined as low-competition. All specifications use county fixed effects, year of origination fixed effects, and loan-level controls including narrow bins for FICO scores, DTI ratios, and LTV ratios, the natural logs of income and loan amount, and indicators for first-time homebuyers, female borrowers, and minority borrowers. Standard errors are clustered at the lender and county level and are reported in parentheses.

Panel B presents tests that examine the effect of litigation exposure on the quality of the representative loan officer serving FHA borrowers in a county. The sample is a county-year panel from 2009 to 2017. The dependent variable in column 1, *Misconduct Rate 2012–2017, (%)*, is the weighted average across lenders, of the fraction of their loan officers with misconduct records, where weights are lenders' FHA market share in the county-year. Lender misconduct rates are computed based on NMLS data covering their loan officers from 2012–2017. The dependent variable in column 2, *Misconduct Rate 2012, (%)*, is similar, but defines lender misconduct rates based on misconduct records already in the NMLS as of 2012. In both cases, lender-level misconduct rates are applied to all of the sample years. The key independent variable is once again the interaction term $Exposure \times Post$. The specifications include the county-year level controls listed and defined in Internet Appendix Table A.1, as well as county fixed effects and year fixed effects. Standard errors are clustered at the county level and are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Table 11

Effect on low-income households' overall access to mortgage credit.

	Low Income Loan Share			Low Income Volume Share		
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure $\times$ Post	-1.160*** (0.428)	-0.093 (0.502)	-1.092** (0.533)	-0.858*** (0.301)	-0.167 (0.348)	-0.864** (0.373)
Exposure $\times$ Post $\times$ Rural		-1.770** (0.780)			-1.146** (0.547)	
Post $\times$ Rural		1.790*** (0.330)			1.153*** (0.222)	
Exposure $\times$ Post $\times$ Minority			-0.031 (0.881)			0.207 (0.630)
Post $\times$ Minority			0.246 (0.350)			0.129 (0.238)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
2010 Mean	10.380	10.380	10.380	5.831	5.831	5.831
Adjusted R ²	0.575	0.577	0.575	0.557	0.559	0.557
# Observations	23,820	23,820	23,820	23,820	23,820	23,820

Note: This table presents difference-in-differences (OLS) regressions examining the effect of the 2012 increase in DOJ litigation risk and large banks' exit from the FHA market on low-income households' total access to mortgage credit across all loan types (FHA and non-FHA). We examine mortgage lending activity from 2009 to 2017 using a county-year panel constructed from HMDA data on all first-lien home purchase mortgages for owner-occupied one-to-four family dwellings. The dependent variable in columns 1–3 is *Low Income Loan Share*, which is defined as the fraction of mortgage loans made to borrowers with incomes below 50% of the FFIEC Median Family Income in the county. Column 1 presents the main test, where the interaction term *Exposure $\times$ Post* is the key independent variable. *Exposure* is defined as the FHA market share of large banks (those among the top 5% largest FHA lenders) as of 2010 in a given county. *Post* is an indicator for the year being 2012 or later. Columns 2 and 3 present tests that include further interactions with *Rural*, an indicator for the county being considered rural/underserved by the CFPB, and with *Minority*, an indicator for the county being in the top tercile of minority population share. All specifications also include the county-year level controls listed and defined in Internet Appendix Table A.1, as well as county fixed effects and year fixed effects. Columns 4–6 present a similar set of tests, where the low-income share of mortgage lending is measured based on dollar volumes rather than loan counts. Standard errors are clustered at the county level and are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

5.10 Table 10: pricing and service quality

Read:

- Table 10 asks whether borrowers benefited through lower rates or better loan-officer quality.
- Panel A finds no significant rate decline in the full sample. In low-competition counties, exposure is associated with a roughly **6 basis point** increase in FHA mortgage rates.
- Panel B finds that loan officer misconduct rates increase in high-exposure counties.
- A one-standard-deviation increase in exposure corresponds to about a **2.5 basis point** increase in misconduct rates, roughly **10%** of the unconditional mean.
- The evidence suggests that smaller shadow banks partly filled the lending gap, but not with better measured service quality.

5.11 Table 11 and Section 7: low-income access and post-2019 policy changes

Read:

- Table 11 is the paper's broader credit-access result.
- The main exposure-by-post coefficient for low-income loan share is **-1.160 percentage points**.
- The 2010 mean low-income loan share is **10.38%**, so the effect is economically large, about **11%** of the mean.

- The effect is especially strong in rural/underserved counties: the rural interaction is **-1.770 percentage points** for loan share and **-1.146 percentage points** for volume share.
- The high-minority interaction is not statistically significant.
- Section 7 extends the evidence beyond 2017 and studies 2019 HUD/DOJ policy changes intended to reduce False Claims Act litigation risk. The paper finds no strong evidence that large banks returned to FHA or that low-income mortgage shares recovered through 2023.

6 Short summary

- **Method.** The paper uses lender-level and county-level difference-in-differences and triple-differences designs around the 2012 DOJ False Claims Act litigation wave in FHA lending.
- **Identification.** The key variation is exposure to top 5% FHA bank lenders before the litigation wave, plus comparisons with conventional lending to rule out broad mortgage-market trends.
- **Main result.** Large banks sharply exit FHA lending after litigation risk rises. In counties heavily exposed to these banks, aggregate FHA lending falls by about 18%.
- **Substitution.** Other lenders expand, especially smaller shadow banks, but they replace only about 60% of the large-bank FHA volume decline.
- **Quality.** The reduction in FHA lending is not offset by better underwriting, lower default rates, lower prices, or better loan-officer quality.
- **Distributional effect.** Low-income borrowers lose access to mortgage credit, especially in rural and underserved counties where replacement lenders are less available.

7 Conclusion

7.1 Core conclusion

- The paper shows that litigation risk can materially change lender participation in a government-supported credit market.
- The DOJ False Claims Act litigation wave led large banks to retreat from FHA lending.
- This retreat reduced aggregate FHA lending in exposed counties and reduced total mortgage credit to low-income households.
- The credit contraction did not produce measurable improvements in underwriting standards or default performance.

7.2 Economic intuition

- FHA lending exposes lenders to certification and enforcement risk. If legal liability is large and hard to predict, even profitable lending can become unattractive.
- Large banks have more franchise value and reputation at stake, so they may respond more strongly to legal risk than monoline shadow banks.
- Other lenders can expand after large banks exit, but substitution is incomplete because lender networks, underwriting capacity, servicing infrastructure, local relationships, and compliance costs are not perfectly transferable.
- Low-income borrowers are disproportionately affected because they rely more heavily on FHA loans and have fewer close substitutes in the conventional market.

7.3 Takeaway

This paper is important because it studies enforcement not only as a penalty for past misconduct but also as a shock to future credit supply. The central lesson is that legal penalties can have real distributional consequences when they push major intermediaries out of markets that serve constrained households. In this case, government litigation risk appears to have reduced low-income borrowers' access to mortgage credit without delivering clear improvements in measured FHA loan quality.